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68.     monthName = MONTHS[birthday.month - 1]
69.     dateText = '{} {}'.format(monthName, birthday.day)
70.     print(dateText, end='')
71. print()
72. print()
73.
74. # Determine if there are two birthdays that match.
75. match = getMatch(birthdays)
76.
77. # Display the results:
78. print('In this simulation, ', end='')
79. if match != None:
80.     monthName = MONTHS[match.month - 1]
81.     dateText = '{} {}'.format(monthName, match.day)
82.     print('multiple people have a birthday on', dateText)
83. else:
84.     print('there are no matching birthdays.')
85. print()
86.
87. # Run through 100,000 simulations:
88. print('Generating', numBDays, 'random birthdays 100,000 times...')
89. input('Press Enter to begin...')
90.
91. print('Let\'s run another 100,000 simulations.')
92. simMatch = 0 # How many simulations had matching birthdays in them.
93. for i in range(100_000):
94.     # Report on the progress every 10,000 simulations:
95.     if i % 10_000 == 0:
96.         print(i, 'simulations run...')
97.         birthdays = getBirthdays(numBDays)
98.         if getMatch(birthdays) != None:
99.             simMatch = simMatch + 1
100. print('100,000 simulations run.')
101.
102. # Display simulation results:
103. probability = round(simMatch / 100_000 * 100, 2)
104. print('Out of 100,000 simulations of', numBDays, 'people, there was a')
105. print('matching birthday in that group', simMatch, 'times. This means')
106. print('that', numBDays, 'people have a', probability, '% chance of')
107. print('having a matching birthday in their group.')
108. print('That\'s probably more than you would think!')

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Exploring the Program

Try to find the answers to the following questions. Experiment with some modifications to the code and rerun the program to see what effect the changes have.

1. How are birthdays represented in this program? (Hint: look at line 16.)
2. How could you remove the maximum limit of 100 birthdays the program generates?